

SAHANA BANNIGOL

+44 7379 422526 · sahanasb.bannigol@gmail.com · linkedin.com/in/sahana-bannigol-0870751b1 · github.com/sahana-bannigol

SUMMARY

Full-stack blockchain engineer, spanning smart contracts, backend APIs, and on-chain integration. Built on 2+ years of backend engineering experience, an MSc in FinTech with Distinction from Exeter, and hands-on blockchain work at Nexa.

WORK EXPERIENCE

Software Developer Consultant

Nexa (L1 Blockchain)

Jul 2026 – Present, UK, Remote

- Integrated Nexa's **existing mint-core module** via a new minting plugin, then built the on-chain flow end to end: creating token groups and minting phage, sample, and stage-data records as NFTs via multipart uploads, plus a **confirmation safeguard** before each irreversible mint.
- Wrote **reconciliation checks** that confirm each mint succeeded on-chain, that associated zip files landed on-chain, and that the database stays in sync with on-chain state, alongside request validation and error handling to harden the API.
- Audited GET routes for complete, accurate field coverage, and fixed a **ticker-collision bug** with randomized, retry-safe ticker generation.

Blockchain Developer Intern

Nexa (L1 Blockchain)

Mar 2026 - Jun 2026, UK, Remote

- Ramped up on the Nexa L1 specification, then bootstrapped the **Ktor backend** from scratch (Gradle Kotlin DSL, Routing/Serialization/Config plugins) with initial domain models for Phage, PhageData, and Sample, the phage lifecycle stage definitions, a health-check route, and a first test.
- Built out the relational backend for the Phages platform, Postgres schema across **8 Flyway migrations** covering labs, users, samples, phages, and phage data, **full CRUD routes for all 5 entities**, and a dashboard UI for data entry with JSON-upload auto-population.

Software Engineer

MapleLabs Solutions

Aug 2022 - Aug 2024, Bengaluru, India

- Built and maintained production Go and Python backend services, including fault-tolerant Kafka telemetry pipelines and diagnostic tooling that **cut monitoring overhead by 25%** and supported resolution of **40+ high-severity incidents** while maintaining SLAs.
- Designed SnappyFlow logging and monitoring plugin integrations across multiple platforms and wrote Go plugins detecting configuration drift against Cisco Intersight.
- Delivered a solo Microsoft Sentinel integration in **5 weeks**: an Azure Function for API extraction, with results visualized in custom KQL dashboards.
- Delivered a solo POC that **won a new enterprise contract**, remediated a critical security vulnerability found in a client audit, **mentored 3 junior engineers**, and collaborated cross-functionally on architecture reviews.

KEY PROJECTS

GreenTrace: Tokenised Green Bond Platform · UKFinnovator+ Hackathon, Imperial College London

Jun 2026

- Built bond issuance and assurance infrastructure on the XRP Ledger using SHA-256 crypto-condition escrow for atomic T+0 settlement, targeting the **\$2.9T green bond market**, plus a V2 multi-agent AI compliance oracle for continuous on-chain verification. Repo: [Link](#) · Demo: [GreenTrace](#)

G-LOC: Digitalised Letters of Credit · MSc Thesis, Distinction

May 2025

- Deployed Solidity smart contracts automating the Letter of Credit lifecycle, cutting B2B settlement from 10 to 14 days to under 24 hours.
- Applied re-entrancy guards, role-based access control, and gas optimization throughout the contract suite.
- Built a JavaScript/MetaMask integration with multisig verification and IPFS storage, backed by 11+ Hardhat tests at 100% coverage. Repo: [Link](#) · Demo: [G-LOC](#)

ExeFit: Gym Membership & ERC-20 Token Reward Platform · Personal Project

Mar 2025

- Built a blockchain-powered gym membership platform with MetaMask non-custodial wallet integration for transaction signing and on-chain reward distribution. Repo: [Link](#) · Demo: [ExeFit](#)

SKILLS

Backend & Distributed Systems: Golang, Python, Kotlin/Ktor, Django, REST API Design, Microservices, Kafka, PostgreSQL, SQL

Blockchain: Solidity, Ethereum, Hardhat, ERC-20/721, Nexa L1, XRPL (XLS-33 MPT, XLS-70 Credentials), IPFS, SHA-256/Double SHA-256, Crypto-condition Escrow, On-chain Auditability

Cloud, DevOps & Observability: AWS, Azure Functions, Kubernetes, Docker, CI/CD Pipelines, GitHub/GitLab, SnappyFlow, Microsoft Sentinel, Fluentbit

Frontend & AI-Assisted Development: TypeScript, JavaScript, React.js, Custodial & Non-custodial Wallet Integration; Claude Code, GitHub Copilot, Gemini, AI-Assisted Test Generation

CERTIFICATIONS

Smart Contracts (Solidity), Satolearn; Go Essentials: Concurrency & High-Performance Applications.

EDUCATION

MSc Financial Technology, Distinction · University of Exeter, UK

Sep 2024 to Sep 2025

B.E. Computer Science & Engineering, Distinction · Vemana Institute of Technology, India

Aug 2018 to Jul 2022